A Self-Supervised Deep Denoiser for Hyperspectral and Multispectral Image Fusion

Zhicheng Wang, Michael K. Ng, *Senior Member, IEEE*, Joseph Michalski, and Lina Zhuang, *Member, IEEE*

***Abstract*— The plug-and-play (PnP) technique enables us to plug image priors into an alternating direction method of multipliers (ADMM) framework for solving a regularized optimization problem. Deep image priors have shown their flexibility and robustness in solving several image inverse problems. Hyperspectral image (HSI) super-resolution problem is an ill-posed inverse problem that aims to obtain a high-resolution HSI (HR-HSI) by combining the information of low-resolution HSI (LR-HSI) and HR multispectral image (HR-MSI) simultaneously. This article proposes an HSI and MSI fusion framework termed E2E-Fusion, plugged with a self-supervised deep learning prior called *Eigenimage2Eigenimage*. First, the spectral low-rank structure of HSIs is exploited via subspace representations of spectra vectors. Meanwhile, benefiting from the high quality of the first eigenimage (i.e., representation coefficients), we design a self-supervised deep eigenimage guidance network image prior, E2E. By using the PnP technique, we plugged the E2E prior into the ADMM fusion framework to update the optimal objective function iteratively. The numerical experimental results both on the simulated datasets and real datasets demonstrate that the proposed method performs better than state-of-the-art fusion methods.**

***Index Terms*— Deep plug-and-play (PnP) prior, hyperspectral image (HSI), image fusion, self-supervised learning.**

I. Introduction

HYPERSPECTRAL imaging technique plays a vital role in Earth observation and deep space exploration [1], [2]. Hyperspectral images (HSIs) are with hundreds of bands, ranging from visible to thermal infrared, which can identify the materials precisely [3], [4], [5]. Due to the limitation of hyperspectral cameras [6], the HSIs usually have a lower spatial resolution than RGB images, panchromatic images, and multispectral images (MSIs). The image fusion technique provides a potential solution to obtain high-resolution HSIs (HR-HSIs) by combining low-resolution HSIs (LR-HSIs) and high-resolution MSIs (HR-MSIs).

HSI and MSI image fusion methods can be mainly divided into two categories, including traditional machine learning-based methods and deep learning-based methods. For the first category, pan-sharpening-based methods and various matrix/tensor factorization methods were proposed to obtain the HR-HSIs. Pan sharpening aims to fuse a low-resolution MSI (LR-MSI) and a high-resolution panchromatic image, which was extended into the HSI–MSI fusion method [7], [8]. Considering the inherent texture of the data, a series of priors were explored, for example, sparsity in [9], [10], and [11], low-rank representations in [12], [13], and [14], and total variation (TV) in [15]. Meanwhile, some HSI and MSI fusion methods were proposed based on the spectral unmixing theory [1], [16]. For example, a coupled nonnegative matrix factorization (CNMF) unmixing method is proposed in [17] for HSI–MSI fusion. More specifically, the HR-HSI is reconstructed based on the endmembers obtained from the LR-HSI and the abundance maps from the HR-MSI, respectively. Tensor-based methods can process the HSI data directly and draw much attention [18], [19], [20], [21], [22], [23]. For instance, a nonlocal sparse tensor factorization method based on Tucker decomposition was proposed in [24], which factorizes an HSI into a sparse core tensor multiplication by dictionaries along the spatial dimension and spectral dimension. In recently, deep learning has been applied in the remote sensing community successfully [25], [26], [27], [28]. More specifically, supervised learning trains a model using labeled data, i.e., noisy–clean pairs. Compared with supervised learning, unsupervised learning trains a model from unlabeled data to explore the information of the data itself. Self-supervised learning is a special kind of unsupervised learning. By generating pretext labels according to the specific tasks, the model trains itself similar to the supervised learning way to learn one part of the input from that of another part, i.e., paired subimages from a single noisy image. The deep learning-based HSI and MSI fusion methods boost rapidly and obtain promising performance. The deep fusion methods can be mainly divided into two categories: supervised learning and unsupervised learning [29], [30], [31], [32], [33], [34], [35], [36], [37]. In the supervised learning methods, the HR-HSIs are needed to train the network, but they are unavailable in the real scene [38]. Accordingly, some unsupervised HSI–MSI

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methods are proposed in [39], [40], and [41] to avoid the requirement of HR-HSI for training.

The plug-and-play (PnP) technique is a flexible framework and can be plugged into alternating direction method of multipliers (ADMM) for solving a regularized optimization problem [42]. Plenty of works have shown that a denoiser can implicitly serve as the image prior for model-based methods to solve many hyperspectral inverse problems, such as image denoising [43], image inpainting [44], compressive sensing [45], super-resolution [46], [47], spectral unmixing [48], and hyperspectral anomaly detection [49]. The PnP technique enables us to use a state-of-the-art denoising operator as an image prior in an optimization model to improve fusion performance. Therefore, this article introduces a self-supervised deep image prior, called *Eigenimage2Eigenimage* (E2E), for HSI–MSI fusion. First, we exploit the fact that spectral vectors of the clean HSI usually live in a low-dimensional orthogonal subspace [50]. To solve the problem of lack ideal training data in real world, this work resorts to the self-supervised learning technique to train the network, which means the model trains itself to learn the mapping only from the noisy eigenimages. Benefiting from the images of subspace coefficients, term *eigenimages*, that have very high spatial correlation and high quality of the first eigenimage, we design an eigenimage-guided deep image prior. Contributions of this work are summarized as follows.

1) We propose a new HSI and MSI fusion method that exploits the spectral low rankness of the underlying HR-HSI by representing its spectral vectors in a prelearned subspace. Consequently, the fusion problem is turned into an eigenimage estimation problem. Instead of designing an elaborate handcrafted regularization for eigenimage estimation, our work resorts to a PnP technique, that is, to plug a deep eigenimage prior into our fusion framework.
2) To learn the inherent data structure of eigenimages effectively, a self-supervised eigenimage learning network, termed E2E, is trained without the requirement of clean data. The E2E prior explores two main characteristics of eigenimage data: 1) neighboring pixels in eigenimages are highly correlated; therefore, we can generate two noisy paired subimages from a single noisy image for network training and 2) first eigenimage is high quality with high signal-to-noise ratios (SNRs); we use first eigenimage (corresponding to the largest eigenvalue) as an auxiliary input and design an eigenimage-guided network.
3) The performance of most HSI and MSI fusion techniques, in particular traditional machine learning-based models proposed in the literature, is sensitive to the setting of regularization parameters. To solve this, we propose a fusion method that is user-friendly in the sense that the only parameter, subspace dimension, may be learned from observed images automatically. Furthermore, the plugged eigenimage network is blind, requiring no parameter.

The rest of this article is as follows. In Section II, we list some related works that inspire our work. We formulate the problem and describe the proposed E2E-Fusion method in Section III. Section IV presents experimental results both on simulated datasets and real datasets that demonstrate the performance of the proposed method. Finally, the conclusion is given in Section V.

II. Related Work

A. Deep PnP Prior

The performance of supervised learning methods heavily depends on the training data, which should be a large amount of well-paired LR-HSI and HR-HSI images, which is infeasible to apply in the practice scene. In addition, in unsupervised learning methods, the designed model must be retained when dealing with a new image, which lacks flexibility and robustness compared with the traditional machine learning methods. To take the advantage of the traditional machine model and deep network simultaneously, effective solutions using deep image priors were proposed in [51] and [52]. In addition, when solving a regularized optimization problem using ADMM, a subproblem composed of a data fidelity term and a regularization term is the so-called "Moreau proximal operator" or "denoising operator"; hence, state-of-the-art deep networks can be successfully used to solve inverse problems of images. In [46], a fast and flexible denoising convolutional neural network (FFDNet) [53] well trained with paired (noisy–clean) realistic images was plugged into the HSI–MSI fusion framework. However, in the HSI remote sensing field, it is challenging to collect plenty of well-paired data. To solve this problem, researchers explore the possibility of training deep learning-based denoisers only with noisy images. We refer to some representative works: Noise2Noise [54], Noise2void [55], Noise2Self [56], and Neighbor2Neighbor [57]. This situation motivates us to study a self-supervised deep PnP network, which can fully use the extremely high spectral correlation of HSIs.

III. Proposed Blind E2E-Fusion Method

We propose a novel user-friendly HSI and MSI fusion method plugged with a blind deep self-supervised image prior. By exploiting two characteristics of eigenimages, we propose E2E-Fusion, an HSI and MSI fusion framework via PnP technique by using the deep self-supervised prior.

A. Problem Formulation

Let a third-order tensor $\mathcal{Y}_h \in \mathbb{R}^{n_{\text{row}} \times n_{\text{col}} \times L}$ denote an LR-HSI with $n_{\text{row}} \times n_{\text{col}}$ spectral vectors and L spectral bands, acquired by a hyperspectral sensor. Assuming that the noise in the image follows a Gaussian distribution, the observed LR-HSI can be written as follows:

$$\mathcal{Y}_h = \mathcal{X}_h + \mathcal{N}_h \tag{1}$$

where $\mathcal{X}_h \in \mathbb{R}^{n_{\text{row}} \times n_{\text{col}} \times L}$ and $\mathcal{N}_h \in \mathbb{R}^{n_{\text{row}} \times n_{\text{col}} \times L}$ denote the clean LR-HSI cube and the corresponding white Gaussian noise cube, respectively.

The HR-MSI is acquired by a multispectral sensor, which can also be expressed as the third-order tensor data, i.e., $\mathcal{Y}_m \in \mathbb{R}^{N_{\text{row}} \times N_{\text{col}} \times l}$, with the size of $N_{\text{row}} \times N_{\text{col}}$ and l spectral bands.

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Assuming that the noise in the image follows a Gaussian distribution, the observed HR-MSI can be defined as follows:

$$\mathcal{Y}_m = \mathcal{X}_m + \mathcal{N}_m \tag{2}$$

where $\mathcal{X}_m \in \mathbb{R}^{N_{\text{row}} \times N_{\text{col}} \times l}$ and $\mathcal{N}_m \in \mathbb{R}^{N_{\text{row}} \times N_{\text{col}} \times l}$ denote the clean HR-MSI cube and the corresponding white Gaussian noise cube, respectively.

Our aim is to obtain an HR-HSI, $\mathcal{Z} \in \mathbb{R}^{N_{\text{row}} \times N_{\text{col}} \times L}$, by making full use of the spectral information of LR-MSI and the spatial information of HR-MSI in the same scene simultaneously. Among them, $N_{\text{row}} > n_{\text{row}}$, $N_{\text{col}} > n_{\text{col}}$, and $L > l$.

In this article, we unfold the data cubes of LR-HSI, HR-MSI, and HR-HSI into matrix format along the spectral dimension, termed $\mathbf{Y}_h \in \mathbb{R}^{L \times n_{\text{row}} n_{\text{col}}}$, $\mathbf{Y}_m \in \mathbb{R}^{l \times N_{\text{row}} N_{\text{col}}}$, and $\mathbf{Z} \in \mathbb{R}^{L \times N_{\text{row}} N_{\text{col}}}$, respectively. LR-HSI can be regarded as a reduced quality representation of HR-HSI in the spatial dimension; hence, the LR-HSI can be expressed as follows:

$$\mathbf{Y}_h = \mathbf{ZBS} + \mathbf{N}_h \tag{3}$$

where $\mathbf{B} \in \mathbb{R}^{N_{\text{row}} N_{\text{col}} \times N_{\text{row}} N_{\text{col}}}$ is a spatial blurring operator of $\mathbf{Z}$ and represents the hyperspectral sensor's point spread function (PSF). Matrices $\mathbf{S} \in \mathbb{R}^{N_{\text{row}} N_{\text{col}} \times n_{\text{row}} n_{\text{col}}}$ and $\mathbf{N}_h \in \mathbb{R}^{L \times n_{\text{row}} n_{\text{col}}}$ are the spatial subsampler and additive Gaussian noise, respectively. The blurring operator $\mathbf{B}$ is detailed as follows:

$$\mathbf{B} = \mathbf{FKF}^{-1} \tag{4}$$

where $\mathbf{F}$ and $\mathbf{F}^{-1}$ represent the fast Fourier transform (FFT) and its inverse transformation. $\mathbf{K}$ is a diagonal matrix that holds the eigenvalues of the matrix $\mathbf{B}$.

Accordingly, the HR-MSI can be regarded as a downsampled version of the HR-HSI in spectral dimension, which can be represented as follows:

$$\mathbf{Y}_m = \mathbf{MZ} + \mathbf{N}_m \tag{5}$$

where $\mathbf{M} \in \mathbb{R}^{l \times L}$ and $\mathbf{N}_m \in \mathbb{R}^{l \times N_{\text{row}} N_{\text{col}}}$ are the spectral response function and the white Gaussian noise, respectively.

We assume that spectral vectors of LR-HSIs live in a low-dimensional subspace because of the strong correlation among the bands [44], [50], [58], [59]. The subspace representation of HSIs has been widely used in HSI processing fields and has shown promising results [60]. Hence, HR-HSI, $\mathcal{Z} \in \mathbb{R}^{N_{\text{row}} \times N_{\text{col}} \times L}$, can also be represented into a low-dimensional subspace. Based on the subspace theory, the $\mathbf{Z}$ can be expressed as follows:

$$\mathbf{Z} = \mathbf{CA} \tag{6}$$

where columns of $\mathbf{C} \in \mathbb{R}^{L \times p} (p \ll L)$ hold an orthogonal basis for the p-dimensional subspace and $\mathbf{C}^T\mathbf{C} = \mathbf{I}_k$ with $\mathbf{I}_k$ representing the identity matrix of dimension k. Matrix $\mathbf{A} \in \mathbb{R}^{p \times N_{\text{row}} N_{\text{col}}}$ holds the representation coefficients of $\mathbf{O}$ with respect to $\mathbf{C}$, and we will call the images associated with rows of $\mathbf{A}$ as *eigenimages*. Furthermore, the dimension of the subspace, p, is much smaller than L. As the LR-HSI can be regarded as a downsampled version of an HR-HSI in the spatial domain, hence, we assume that the LR-HSI and HR-HSI share the same subspace $\mathbf{C}$ [15], [46].

Hence, the spectral subspace of $\mathbf{C}$ can be estimated via eigendecomposition of the spectral correlation matrix [denoted as $\mathbf{R}_y = \mathbf{Y}_h\mathbf{Y}_h^T/(n_{\text{row}} n_{\text{col}})$] of the LR-HSI

$$\mathbf{R}_y = \mathbf{U}\boldsymbol{\Sigma}^2\mathbf{U}^T \tag{7}$$

where $\mathbf{U}$ and $\boldsymbol{\Sigma}$ are eigenvector and eigenvalue matrices of the correlation matrix $\mathbf{R}_y$, respectively. The eigenvalues in diagonal matrix $\boldsymbol{\Sigma}^2$ are ordered by nonincreasing magnitude. The spectral subspace is approximately spanned by p eigenvectors associated with the p largest eigenvalues, i.e.,

$$\mathbf{C} = \mathbf{U}(:, 1:p). \tag{8}$$

Using subspace representation of spectral vectors, observation models of LR-HSI and HS-MSI can be reformulated as follows:

$$\mathbf{Y}_h = \mathbf{CABS} + \mathbf{N}_h \tag{9}$$

and

$$\mathbf{Y}_m = \mathbf{MCA} + \mathbf{N}_m. \tag{10}$$

B. Fusion Method: E2E-Fusion

Given the spectral subspace $\mathbf{C}$, the fusion of the LR-HSI and HR-HSI can be formulated as follows:

$$\min_{\mathbf{A}} \frac{1}{2}\|\mathbf{Y_h} - \mathbf{CABS}\|_F^2 + \frac{\lambda_1}{2}\|\mathbf{Y_m} - \mathbf{MCA}\|_F^2 + \lambda_2\phi_{\text{E2E}}(\mathbf{A}) \tag{11}$$

where first two terms are data fidelity derived from observation models (9) and (10), respectively. Nonnegative parameter λ_1 models the noise intensity difference between two images in terms of Frobenius norm. Symbol λ_2 is nonnegative regularization parameter, whose setting depends on the noise intensity of $\mathbf{A}$. The function $\phi_{\text{E2E}}(\cdot)$ represents an image prior extracted from an E2E network. The prior information of the $\mathbf{A}$ has been exploited for LR-HSI and HR-MSI images fusion. Different image priors (such as sparsity, TV, and low-rank priors) have been used on $\mathbf{A}$ to improve the accuracy of the coefficient estimation [9], [15]. In this article, we propose a new self-supervised deep learning image prior, E2E, to estimate the coefficient $\mathbf{A}$. The flowchart of the proposed fusion method is illustrated in Fig. 1.

We solve the objective function (11) via the ADMM. By introducing a multiple auxiliary variable $\mathbf{V}$, the augmented Lagrangian function of (11) is

$$L(\mathbf{A}, \mathbf{V}, \mathbf{G}) = \frac{1}{2}\|\mathbf{Y}_h - \mathbf{CABS}\|_F^2 + \frac{\lambda_1}{2}\|\mathbf{Y}_m - \mathbf{MCA}\|_F^2 + \lambda_2\phi_{\text{E2E}}(\mathbf{V}) + \frac{\mu}{2}\left\|\mathbf{V} - \mathbf{A} + \frac{\mathbf{G}}{\mu}\right\|_F^2 \tag{12}$$

where $\mathbf{G}$ is the Lagrange multiplier and μ is the penalty parameter. The variables $\mathbf{A}$, $\mathbf{V}$, and $\mathbf{G}$ are updated sequentially, and these steps are shown in Algorithm 1. The optimization details of the loss function are as given in the following.

Update A: The optimization problem for $\mathbf{A}$ is

$$\mathbf{A} = \arg\min_{\mathbf{A}} \frac{1}{2}\|\mathbf{Y}_h - \mathbf{CABS}\|_F^2$$

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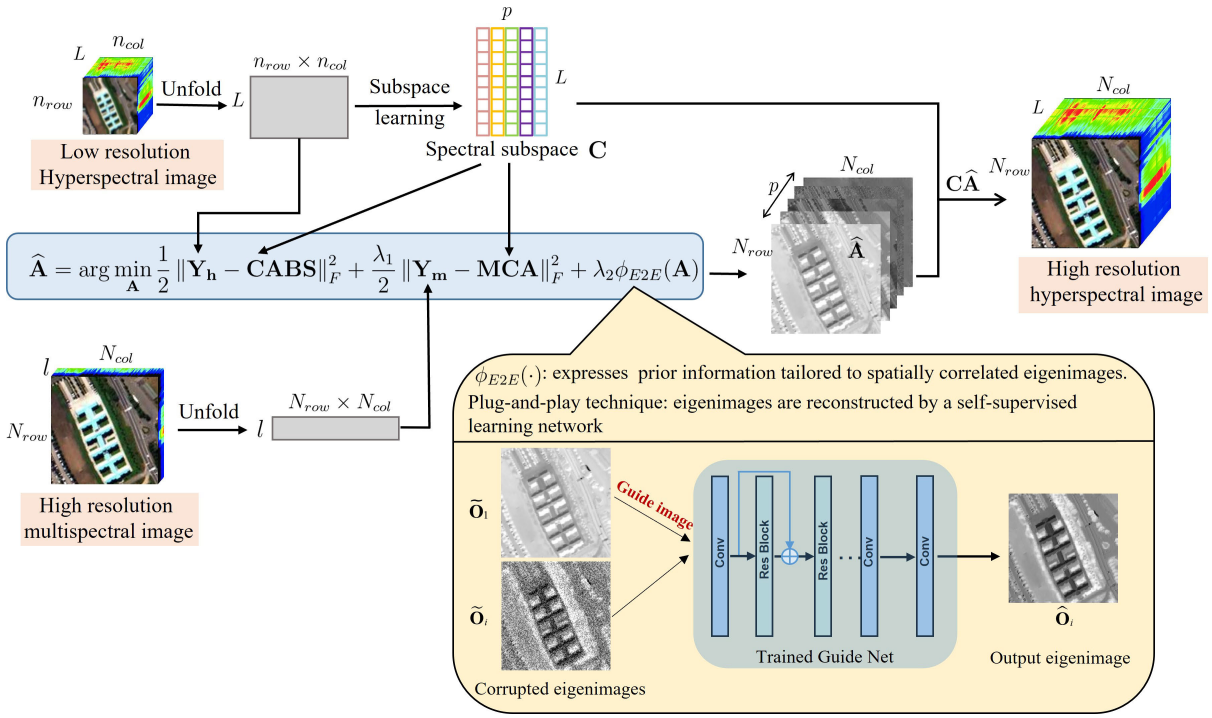


Fig. 1. Flowchart of the proposed fusion framework, E2E-Fusion.

$$+\frac{\lambda_1}{2}\|\mathbf{Y}_m - \mathbf{MCA}\|_F^2 + \frac{\mu}{2}\left\|\mathbf{V} - \mathbf{A} + \frac{\mathbf{G}}{\mu}\right\|_F^2 \tag{13}$$

which is strongly convex. Let the derivative of (13) with respect to $\mathbf{A}$ to be 0; then, we acquire the Sylvester equation (see [61] for more details) as follows:

$$\mathbf{H}_1\mathbf{A} + \mathbf{A}\mathbf{H}_2 = \mathbf{H}_3 \tag{14}$$

where

$$\mathbf{H}_1 = \mu(\mathbf{MC})^T\mathbf{MC} + \mu\mathbf{I} \tag{15}$$

$$\mathbf{H}_2 = \mathbf{BS}(\mathbf{BS})^T \tag{16}$$

and

$$\mathbf{H}_3 = \mathbf{C}^T\mathbf{Y}_h(\mathbf{BS})^T + \lambda_1(\mathbf{MC})^T\mathbf{Y}_m + \mu\left(\mathbf{V} + \frac{\mathbf{G}}{\mu}\right). \tag{17}$$

Update V: The optimization problem for $\mathbf{V}$ is

$$\mathbf{V} = \arg\min_{\mathbf{V}} \frac{\mu}{2}\left\|\mathbf{V} - \mathbf{A} + \frac{\mathbf{G}}{\mu}\right\|_F^2 + \lambda_2\phi_{\mathrm{E2E}}(\mathbf{V}) = \psi_{\lambda\phi_{\mathrm{E2E}}}\left(\mathbf{A} - \frac{\mathbf{G}}{\mu}\right) \tag{18}$$

where $\psi_{\lambda\phi_{\mathrm{E2E}}}(\cdot)$ is the so-called Moreau proximity operator or denoising operator [62] of ϕ_{E2E}. We solve (18) using the PnP technique. The core idea of PnP is to plug an off-the-shelf denoiser into the framework instead of investing effort in designing a new regularizer. In this article, we plug a blind self-supervised deep learning network, E2E [63], into (18), leading to

$$\mathbf{V} \leftarrow \mathrm{E2E}\left(\mathbf{A} - \frac{\mathbf{G}}{\mu}\right) \tag{19}$$

Algorithm 1 Proposed E2E-Fusion Algorithm

Input: LR-HSI:$\mathbf{Y}_h$; HR-MSI:$\mathbf{Y}_m$;
Output: HR-HSI: $\mathbf{Z}$

```
for k = 1; k < Iter; k++ do
    Update A via (14);
    Update V via (19);
    Update G via (20);
    k = k + 1;
end
Z = CA;
return Z
```

where the function E2E(·) outputs denoised eigenimages. A detailed description of the E2E network is given in Section III-C.

Update G: The optimization problem for $\mathbf{G}$ is

$$\mathbf{G} = \mathbf{G} + \mu(\mathbf{V} - \mathbf{A}). \tag{20}$$

C. Plugged Self-Supervised Deep Learning Prior: E2E

The PnP trick enables us to directly use eigenimage priors extracted from deep denoising neural networks. However, the performance of deep learning-based denoisers highly depends on the quantity and quality of training data. Furthermore, paired noisy–clean training eigenimages are generally unavailable in hyperspectral remote sensing areas. To solve this problem, this work resorts to the self-supervised learning technique, meaning our network can train itself to learn one part of noisy input from another part of the noisy input.

Here, we propose a self-supervised deep image prior, termed E2E. This E2E prior explores two main characteristics of

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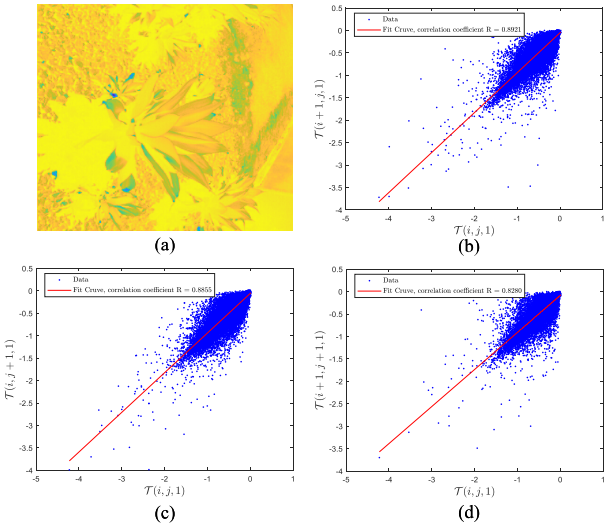


Fig. 2. ICVL image. (a) First eigenimage. (b)–(d) Correlation of two vertical, horizontal, and diagonal adjacent pixels in the first eigenimage.

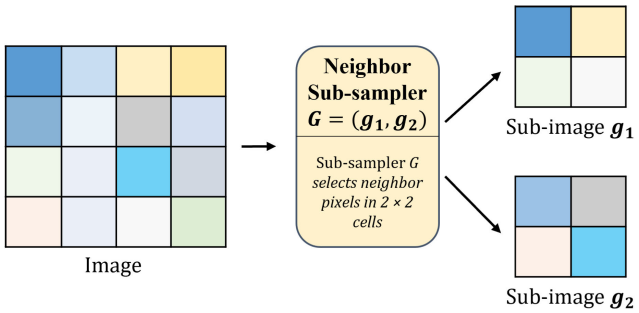


Fig. 3. Toy example of generating an image pair with a neighbor subsampler $G = (g_1, g_2)$.

eigenimage data. First, neighboring pixels in eigenimages $\mathcal{T}$ are highly correlated; therefore, we can generate two noisy paired subimages from a single noisy image for network training. The highly correlated property of eigenimages can be illustrated using a toy example, and one of the Interdisciplinary Computational Vision Laboratory (ICVL) images of size 256 × 256 pixels was carried out. Fig. 2 briefly shows the high correlation between neighboring pixels in the first band of the eigenimage. Fig. 2(a) shows the first band of eigenimage. Correlation coefficients of two horizontal, vertical, and adjacent diagonal pixels in the eigenimage are 0.8912, 0.8855, and 0.8280, respectively [shown in Fig. 2(b)–(d)]. In this article, we further study the high quality of the first eigenimage. More specifically, the first band of eigenimages learned from a noisy HSI has a much higher SNR than the remaining eigenimages, which can be used as a guide image to denoise of remaining eigenimages. The high quality of the first eigenimage can be illustrated using an example. The Washington DC Mall data with the size of 304 × 304 × 191 and the Indian Pines data with the size of 144 × 144 × 191 were used to show the characteristics of the eigenimages. We first project these two data into the low-dimensional subspace, and the number of subspaces, p, is set as 6. Figs. 4(a)–(f) and 5(a)–(f) display the eigenimages in order, respectively. It is evident that the first eigenimage in both datasets exhibits high visual quality. What stands out in Figs. 4(g) and 5(g) are the rapid decrease of the eigenimages in terms of PSNR, and the first eigenimage has a much higher PSNR than the remaining eigenimages on both of two datasets. Hence, based on the phenomenon that the first band of eigenimages has higher quality, we will use it as the guide image to assist in the denoising of other eigenimages.

To use the proposed E2E self-supervised deep prior, we reformulate the matrix $[\mathbf{A} - (\mathbf{G}/\mu)]$ into a third tensor, denoted as $\widetilde{\mathcal{O}} \in \mathbb{R}^{N_{\text{row}} \times N_{\text{col}} \times p}$. For simplicity, let $\tilde{\mathbf{o}}_i$ represent the ith eigenimage, i.e., the ith mode-3 slice of $\widetilde{\mathcal{O}}$. Hence, the solution of (18) can be reformulated as follows:

$$\mathbf{V}(i,:) \leftarrow \text{reshape}(f_\theta(\text{cat}(\tilde{\mathbf{o}}_1, \tilde{\mathbf{o}}_i)), [1, N_{\text{row}} \times N_{\text{col}}]), \quad (i = 1, \ldots, p) \tag{21}$$

where function reshape$(\mathbf{R}, [1, N])$ reshapes matrix $\mathbf{R}$ into a 1-by-N vector. By exploiting the above two characteristics of eigenimage data, we propose a self-supervised learning eigenimage network

$$\hat{\mathbf{o}}_i = f_\theta(\text{cat}(\tilde{\mathbf{o}}_1, \tilde{\mathbf{o}}_i)), \quad (i = 1, 2, \ldots, p) \tag{22}$$

where f_θ is the denoising network parameterized by θ, function cat$(\mathbf{A}, \mathbf{B})$ concatenates matrix $\mathbf{B}$ to the end of matrix $\mathbf{A}$ along dimension 3, and $\tilde{\mathbf{o}}_i := \widetilde{\mathcal{O}}(:,:,i)$ represents the ith corrupted eigenimage and $\hat{\mathbf{o}}_i$ represents an estimation of the ith eigenimage. Here, $\tilde{\mathbf{o}}_1$ serves as a guide image in the network, as it contains high-quality spatial information. We guide our network using the first eigenimage, so that the network can learn structured information from the first eigenimage. As the scene across different eigenimages are the same, the structured learned information can help to build a denoising network for other eigenimages.

The training part of our E2E framework is introduced below. Considering the fact that collecting a large amount of aligned noisy–clean training data is challenging in the hyperspectral remote sensing areas, we generate training eigenimage pairs from single noisy eigenimages using random neighbor subsamplers defined in [57]. Fig. 3 gives a toy example of generating an image pair with a neighbor subsampler $G = (g_1, g_2)$. Basically, the neighbor subsampler enables us to split a single noisy eigenimage into two noisy subeigenimages. More specifically, the subsampler randomly selects two adjacent elements in a 2-by-2 window and assigns them to two subimages. Meanwhile, we only consider horizontal and vertical adjacent pixels, because the correlation coefficients of horizontal and vertical adjacent pixels are larger than diagonal adjacent pixels (shown in Fig. 3).

After generating sampled image pairs, we can train a network with aligned pairwise noisy–noisy eigenimages

$$\arg\min_\theta \sum_i \mathbb{E}_{(\tilde{\mathbf{o}}_1, \tilde{\mathbf{o}}_i)} \| f_\theta(\text{cat}(g_1(\tilde{\mathbf{o}}_1), g_1(\tilde{\mathbf{o}}_i))) - g_2(\tilde{\mathbf{o}}_i) \|_F^2 \tag{23}$$

where $[g_1(\tilde{\mathbf{o}}_i), g_2(\tilde{\mathbf{o}}_i)]$ is a training pair of noisy–noisy eigenimages.

The sampled images are closely resembled, because their pixels are adjacent in the same noisy source image. However, the underlying clean versions of two sampled images usually differ slightly, implying the denoising result of the network

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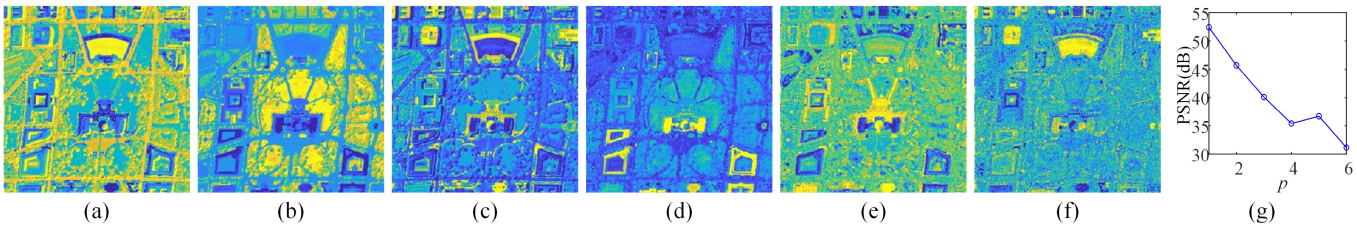


Fig. 4. Washington DC image. (a)–(f) One to six eigenimages. (g) PSNR of each eigenimage.

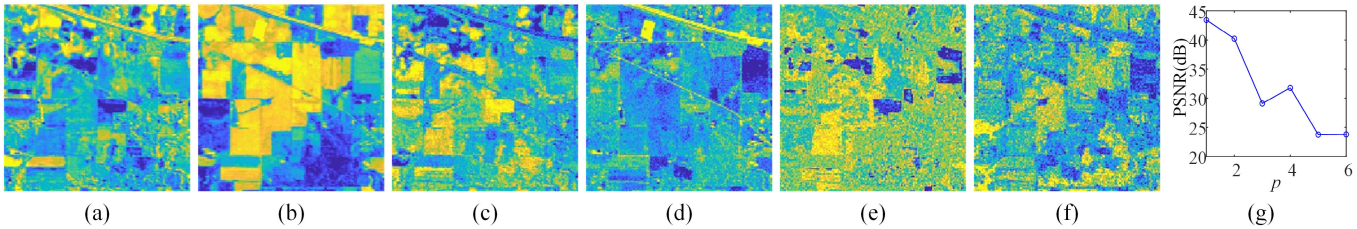


Fig. 5. Indian Pines image. (a)–(f) One to six eigenimages. (g) PSNR of each eigenimage.

optimized by (23) may be over-smooth [57]. To mitigate this over-smoothness, the slight difference between sampled pairs is included in the loss function by adding a constraint below.

By introducing the image difference into (23), we consider the following constrained optimization problem for E2E framework:

$$\begin{aligned}\min_{\theta} \sum_{i} & \mathbb{E}_{(\tilde{\mathbf{o}}_1,\tilde{\mathbf{o}}_i)} \| f_\theta(\text{cat}(g_1(\tilde{\mathbf{o}}_1), g_1(\tilde{\mathbf{o}}_i))) - g_2(\tilde{\mathbf{o}}_i) \|_F^2 \\ & + \gamma \mathbb{E}_{(\tilde{\mathbf{o}}_1,\tilde{\mathbf{o}}_i)} \| f_\theta(\text{cat}(g_1(\tilde{\mathbf{o}}_1), g_1(\tilde{\mathbf{o}}_i))) - g_2(\tilde{\mathbf{o}}_i) \\ & - (g_1(f_\theta(\text{cat}(\tilde{\mathbf{o}}_1, \tilde{\mathbf{o}}_i))) - g_2(f_\theta(\text{cat}(\tilde{\mathbf{o}}_1, \tilde{\mathbf{o}}_i)))) \|_F^2 \end{aligned} \tag{24}$$

where γ is a nonnegative parameter. Compared with the training loss of [57], one can see that in our training loss (24), our network input data is a two-band image (instead of a single-band image in [57]). Our eigenimage denoising network explores the characteristic of the first eigenimage data, which is high quality with high SNRs. Hence, we can use the first eigenimage (corresponding to the largest eigenvalue) as an auxiliary input and design an eigenimage-guided network. Ablation experiments demonstrate that the use of guide eigenimage does improve fusion performance significantly.

D. Fusion of HR-HSI

We estimate eigenimages $\widehat{\mathbf{A}}$ by solving 11. Meanwhile, the subspace basis, $\mathbf{C}$, is obtained by implementing eigendecomposition of the spectral correlation matrix of the LR-HSI. By multiplying $\widehat{\mathbf{A}}$ and C together, then we obtain the HR-HSI, $\mathbf{wide}\widehat{\mathbf{Z}}$

$$\widehat{\mathbf{Z}} = \mathbf{C}\widehat{\mathbf{A}}. \tag{25}$$

IV. Experiments

In this section, we test the effectiveness of the proposed E2E-Fusion method on four simulated datasets and a real dataset, respectively (see Fig. 6 for more details). Meanwhile, we compare with several state-of-the-art methods and analyze the fusion results of all methods visually and quantitatively.

A. Experimental Datasets

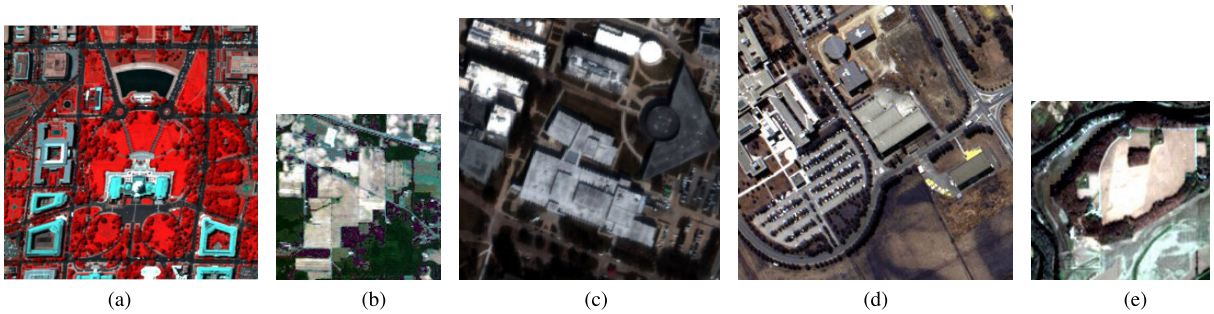
In the study, all simulated datasets were synthesized via a similar method mentioned in [64] and [47]. More specifically, an LR-HSI is synthesized by applying a Gaussian filter with a standard deviation of 2 [46] on an HR-HSI, and the size of the Gaussian kernel is kept the same with the downsampling ratio. An HR-MSI is generated by using the spectral response function from visible to shortwave infrared (SWIR) bands of Landsat 8 [40], [65]. The downsampling ratio is set to 8 for the Washington DC Mall and the University of Houston data and 4 for the Indian Pines and Pavia data. Zero-mean Gaussian white noise was added to clean LR-HSIs and HR-MSIs, yielding a series of noisy images with SNRs of 35 and 30 dB, respectively.

1) Washington DC Mall: The first simulated dataset was widely used in HSI analysis [43], [44], [50]. It was acquired by the HYDICE sensor in 1995 in Washington, DC, USA. This full Washington DC Mall image has a region of 1280×307 pixels, and the spatial resolution is 3 m. The image consists of 210 bands, and spectral wavelength is ranging from 0.4 to 2.5 μm. After removing the low SNRs and atmospheric absorption bands, 191 bands were kept. The subregion consisting of $304 \times 304 \times 191$ pixels was clipped from the full dataset as a reference HR-HSI [see Fig. 6(a)].

2) Indian Pines: The second dataset was acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over the Indian Pines test site in northwestern Indian. The image has 224 spectral bands with the size of 145×145 pixels, and the spectral wavelength range is from 0.4 to 2.5 μm. After removing bands caused by water vapor absorption and low SNRs, the subregion consisting of $144 \times 144 \times 191$ pixels was chosen from the full dataset as a reference HR-HSI [see Fig. 6(b)].

3) University of Houston: The third dataset is University of Houston data, which was used in the 2018 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest [66]. The image contains 601×2384 pixels and

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(a) (b) (c) (d) (e)

Fig. 6. HSIs used in the experiments. (a) Subimage of Washington DC Mall data. (b) Indian Pines. (c) Subregion of the University of Houston data. (d) Subregion of the Pavia University. (e) Sentinel-2A data.

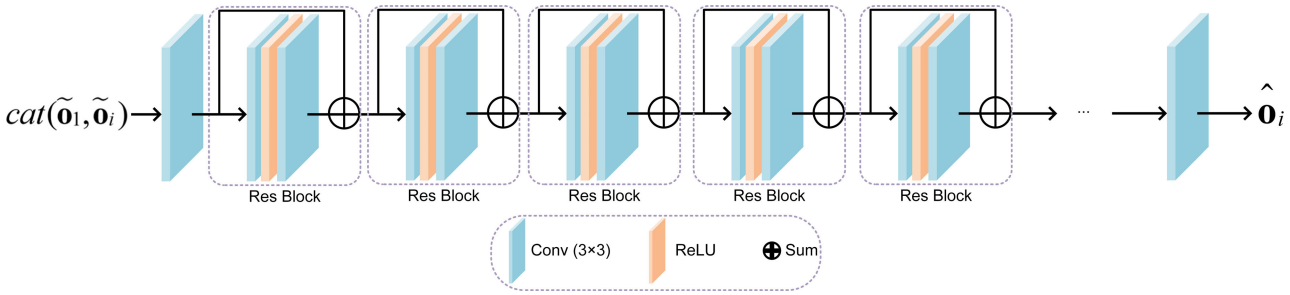


Fig. 7. Details of the proposed guidance network prior, $\mathcal{F}_{\theta \mathrm{E2E}}$.

48 bands ranging from 0.38 to 1.05 μm. The subimage consisting of 320 × 320 × 46 pixels was chosen from the whole dataset as a reference HR-HSI for our experiments [see Fig. 6(c)].

4) Pavia University: The fourth Pavia University data were collected by the reflective optics system imaging spectrometer (ROSIS-3) optical airborne sensor. This image contains 115 bands and with the size of 610 × 340 × 115. After removing bands caused by water vapor absorption and low SNRs, the subregion consisting of 304 × 304 × 191 pixels was clipped from the full dataset as a reference HR-HSI in this study [see Fig. 6(d)].

B. Implementation Details of E2E-Fusion

The proposed E2E-Fusion method is based on a PnP framework, and a ResNet [67] network architecture is used for training the deep PnP prior. More specifically, Fig. 7 displays the proposed network structure in detail. In this article, the network structure includes two basic units: convolution layer and nonlinear activation layer. The $\mathcal{F}_{\theta \mathrm{E2E}}$ is the proposed eigenimage-guided network, which contains R blocks by using two convolution layers and a rectified linear unit (Conv + ReLU + Conv), and the number of kernels and the size were set as 64 and 3 × 3, respectively. Meanwhile, for each block, pixelwised sum was used for aggregating input feature maps and output feature maps. The ICVL hyperspectral dataset[1] [68] was used to train the ResNet network, which was acquired by ground-based cameras with high SNR. Thus, we simulated noisy HSIs by adding independent Gaussian noise into the original ICVL images. Let $\mathbf{n}_i \in \mathbb{R}^b$ denote noise vector of the ith spectral pixel. We simulated pixelwise and bandwise independent Gaussian noise: $\mathbf{n}_i \sim \mathcal{G}(0, \mathbf{D}^2)$, where $\mathcal{G}$ is a zero-mean Gaussian distribution, and $\mathbf{D}$ is a diagonal matrix with diagonal elements $d_{ii} = \alpha \mathrm{median}(\mathcal{X}(:,:,i))$. We set $\alpha = 0.12$. We set the noise intensity of each band according to the median value per band, so that images with diverse noise intensities can be used to train a flexible denoising network. The network training was conducted on the framework of PyTorch. More specifically, the patch size of the image and the batch size are set to 64 and 16, respectively. The initial learning rate of the network is 0.0003 and half-decayed per 20 epochs, and the number of training epochs is 100. The Adam optimizer is chosen for training the network, and the loss function and the value of the hyperparameter in it keep the same value as in Neighbor2Neighbor [57].

C. Compared Methods and Metrics

1) Compared Methods: We compare the proposed method with several state-of-the-art fusion methods, including the generalized Laplacian pyramid hypersharpening (GLPHS) method[2] [6], CNMF[3] [17], hyperspectral superresolution (HySure) method[4] [15], and convolutional neural network based HSI and MSI fusion (CNN-Fus)[5] [46]. It is mentioned that GLPHS, CNMF, HySure, CNN-Fus, and E2E-Fusion methods were tested in MATLAB R2022a on Windows 11 with 12-GB RAM. A demo of MATLAB version for the proposed E2E-Fusion method will be released at http://github.com/LinaZhuang. The parameters in all compared methods (GLPHS, CNMF, HySure, and CNN-Fus) were fine-turned to get the optimal results manually in all

[1] http://icvl.cs.bgu.ac.il/hyperspectral/

[2] http://naotoyokoya.com/Download.html
[3] http://naotoyokoya.com/Download.html
[4] https://github.com/alfaiate/HySure
[5] https://github.com/renweidian/CNN-FUS

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TABLE I
NUMERICAL ANALYSIS OF THE PROPOSED METHOD AND COMPARED METHODS ON FIVE METRICS

Scenario	Metric	GLPHS [6]	CNMF [17]	HySure [15]	CNN-FUS [46]	**E2E-Fusion (Proposed)**
Washington DC	MPSNR (dB)	25.57	26.29	25.15	34.12	**35.57**
	MSSIM	0.8834	0.8754	0.9171	0.9762	**0.9789**
	MFSIM	0.9284	0.9059	0.9372	0.9763	**0.9781**
	SAM	7.8675	7.2016	7.8341	2.9973	**2.8741**
	ERGAS	2.3804	2.2357	2.4975	0.9266	**0.8343**
	Time(s)	38.04	**10.13**	19.09	173.08	328.34
Indian Pines	MPSNR (dB)	25.58	30.13	28.62	31.44	**32.21**
	MSSIM	0.8750	0.9340	0.9272	0.9413	**0.9506**
	MFSIM	0.9375	0.9639	0.9526	0.9670	**0.9692**
	SAM	3.6867	2.6756	3.2686	2.6502	**2.4473**
	ERGAS	3.2974	1.8738	2.1541	1.6736	**1.5513**
	Time(s)	9.08	4.35	**3.30**	50.46	74.65
The University of Huston	MPSNR (dB)	24.15	27.25	26.22	37.65	**38.49**
	MSSIM	0.8548	0.8174	0.8824	0.9690	**0.9782**
	MFSIM	0.9294	0.9183	0.9269	0.9828	**0.9875**
	SAM	6.3338	4.2028	6.9859	2.4635	**1.9229**
	ERGAS	2.5012	1.7542	2.0711	0.6689	**0.6036**
	Time(s)	**2.34**	7.52	17.80	48.32	39.40
Pavia University	MPSNR (dB)	31.88	33.36	32.24	41.94	**42.00**
	MSSIM	0.9178	0.9368	0.9069	0.9809	**0.9818**
	MFSIM	0.9434	0.9645	0.9410	0.9898	**0.9905**
	SAM	4.4292	3.2795	5.6945	2.2050	**2.0889**
	ERGAS	3.8889	3.3437	3.9180	1.2240	**1.2100**
	Time(s)	**10.08**	14.00	20.22	98.30	89.64

experiments. For a fair comparison, the dimension of spectral subspace, p, was set to the same value in the CNN-Fus method and E2E-Fusion method.

2) Quantitative Indexes: We use five well-known metrics to evaluate the image fusion quality numerically when the reference image is available. They are the mean peak SNR (MPSNR) index, the mean structural similarity (MSSIM) index, the mean feature similarity (MFSIM) index, the spectral angle mapper (SAM) index, and the relative dimensionless global error in synthesis (ERGAS) index.

D. Parameter Analysis of the Proposed E2E-Fusion

The dimension of subspace, p, is a crucial parameter, which will influence the fusion results. To test the effect of p, we plot the fusion performance of our method with various of p on four simulated datasets in terms of MPSNR (shown in Fig. 8). It is clear that the fusion results keep a relatively stable value on the Indian Pines image and University of Huston image as p changes. The performance of Pavia University decreased gradually when the p is larger than 7, and the fusion result decreased sharply on the Washington DC Mall. Therefore, the dimension of subspace, p, should be chosen carefully to obtain the best fusion results.

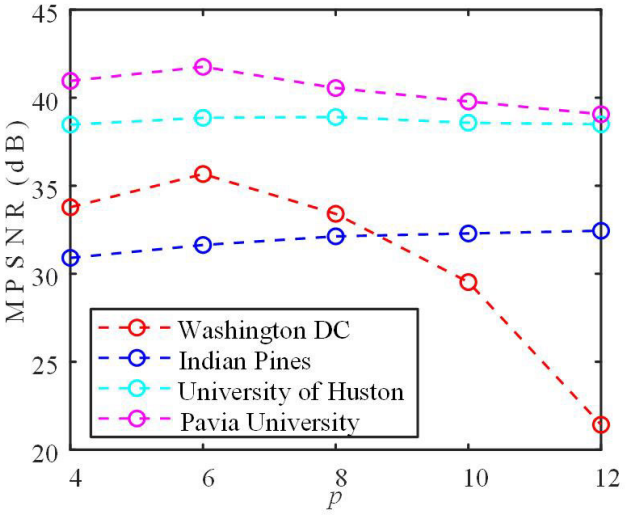


Fig. 8. Effect of the dimension of subspace, p, on four simulated datasets.

E. Experimental Results

1) Experimental Results on Simulated Datasets: The fusion performance of all methods on four synthesized datasets in terms of MPSNR, MSSIM, MFSIM, SAM, and ERGAS is shown in Table I, and the best results are shown in bold. The dimension of spectral subspace, p, was set to 6, 10, 5, and 6 for Washington DC Mall, Indian Pines, University of Huston, and Pavia University, respectively. It is clear that the proposed method can achieve better fusion results in most cases. The proposed E2E-Fusion and CNN-Fus methods perform superior numerically compared with GLPHS, CNMF, and HySure. Both CNN-Fus and E2E-Fusion are PnP-based frameworks. For the CNN-Fus method, an FFDNet with a noise level map was plugged into the fusion framework; therefore, the parameter of the noise level map needs to be fine-tuned carefully to get the optimal results. However, our E2E-Fusion method is user-friendly as a blind, and self-supervised learning network is plugged into our method. Furthermore, a guide image strategy is adopted in our eigenimage denoising network,

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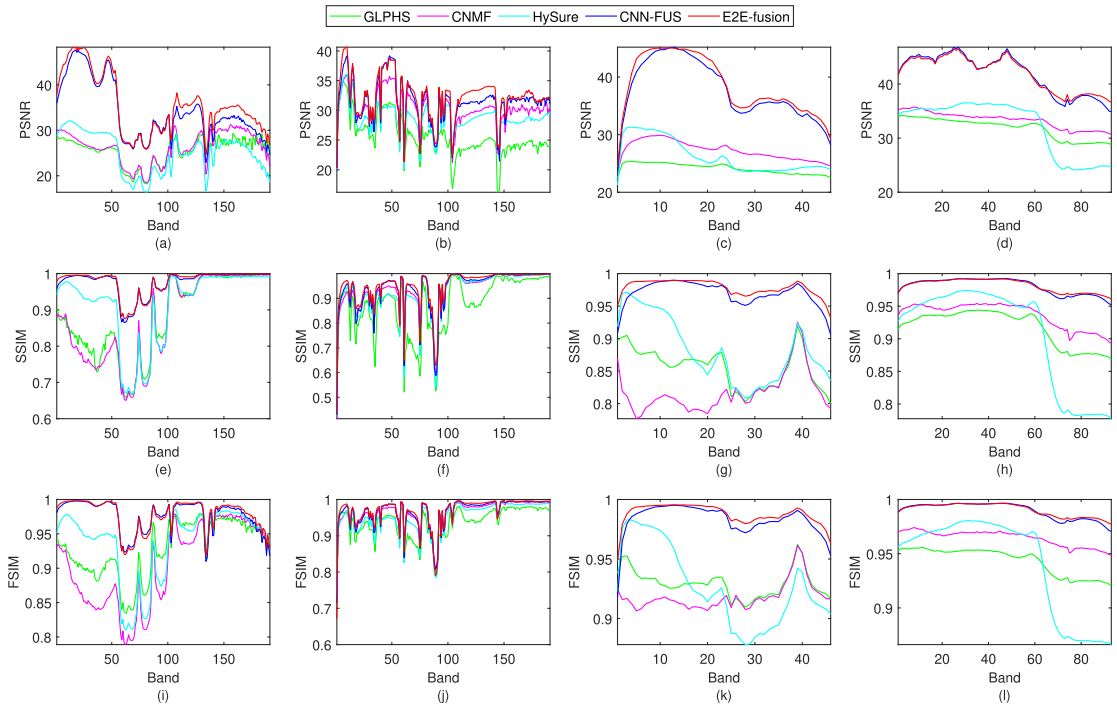


Fig. 9. (a), (e), and (i) PSNR, SSIM, and FSIM value on Washington DC, respectively. (b), (f), and (j) PSNR, SSIM, and FSIM value on Indian Pines, respectively. (c), (g), and (k) PSNR, SSIM, and FSIM value on the University of Huston, respectively. (d), (h), and (l) PSNR, SSIM, and FSIM value on Pavia University, respectively.

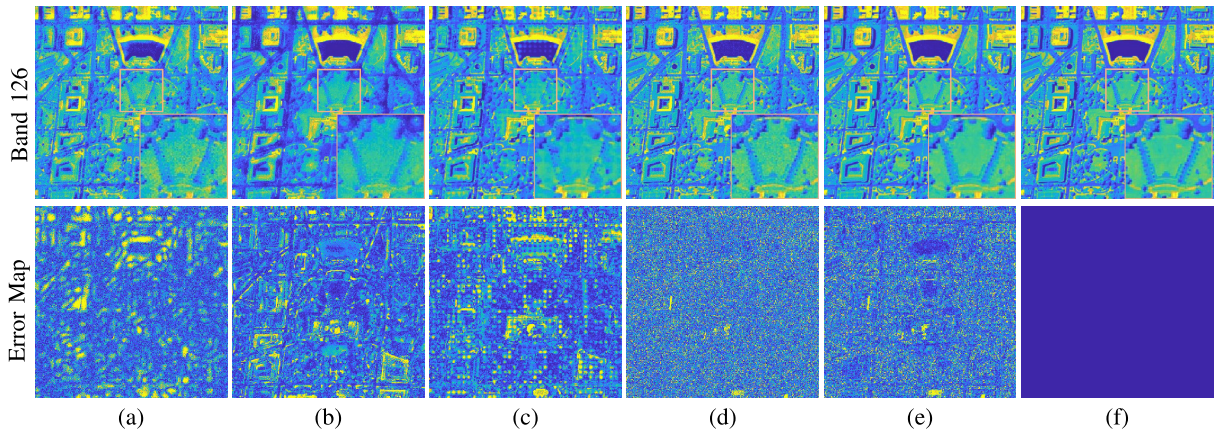


Fig. 10. Band 126, and error map of the Washington DC Mall. (a) GLPHS [6] (GRSM, 2017). (b) CNMF [17] (TGRS, 2012). (c) HySure [15] (TGRS, 2014). (d) CNN-Fus [46] (TNNLS, 2020). (e) E2E-Fusion (Proposed). (f) Reference.

so that the high-quality information of the first eigenimage is fully exploited. Therefore, although our fusion method is blind, its performance is comparable to the unblind CNN-Fus.

The running times of all methods are reported in Table I. For further comparison, we also show the PSNR, SSIM, and FSIM of per band of four simulated datasets in Fig. 9. Meanwhile, we also display the fusion results on four simulated datasets virtually, including the 126th band of Washington DC Mall, the 116th band of Indian Pines, the sixth band of Huston University, and the 90th band of Pavia University. Meanwhile, the corresponding error maps were displayed via the absolute difference values between reference images and fusion results (shown in Figs. 10–13). We can see that the CNN-Fus and the proposed method perform superior over other competitors, and the proposed E2E-Fusion keeps less noise.

We remark that our fusion method is user-friendly, as it plugs a blind eigenimage learning network, requiring no parameter. However, the performance of most HSI and MSI fusion techniques, in particular traditional machine learning-based models proposed in the literature, is sensitive to the setting of regularization parameter. Both our proposed E2E-Fusion and CNN-Fus can achieve a better fusion results compared with other methods. However, to achieve a promising fusion result, the parameter, λ, in CNN-Fus should be turned carefully. What can be clearly seen in Fig. 14, the MPSNR of CNN-Fus is the dramatic decline in the suboptimal parameter range, while the proposed E2E-Fusion (requiring no regularization parameter) can keep a stable fusion accuracy.

2) Ablation Experiments: A series of ablation studies were conducted to verify the effectiveness of our plugged E2E eigenimage prior, and the fusion results are shown in Table II. We first change the objective function into $\min_{\mathbf{A}}(1/2)\|\mathbf{Y_h} - \mathbf{CABS}\|_F^2 + (\lambda_1/2)\|\mathbf{Y_m} - \mathbf{MCA}\|_F^2$, with

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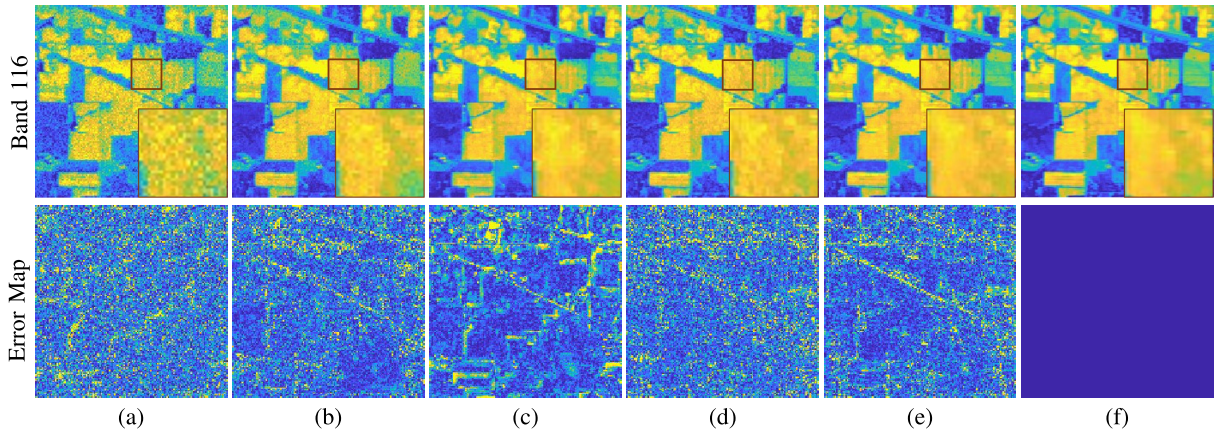


Fig. 11. Band 116, and error map of the Indian pines. (a) GLPHS [6] (GRSM, 2017). (b) CNMF [17] (TGRS, 2012). (c) HySure [15] (TGRS, 2014). (d) CNN-Fus [46] (TNNLS, 2020). (e) E2E-Fusion (Proposed). (f) Reference.

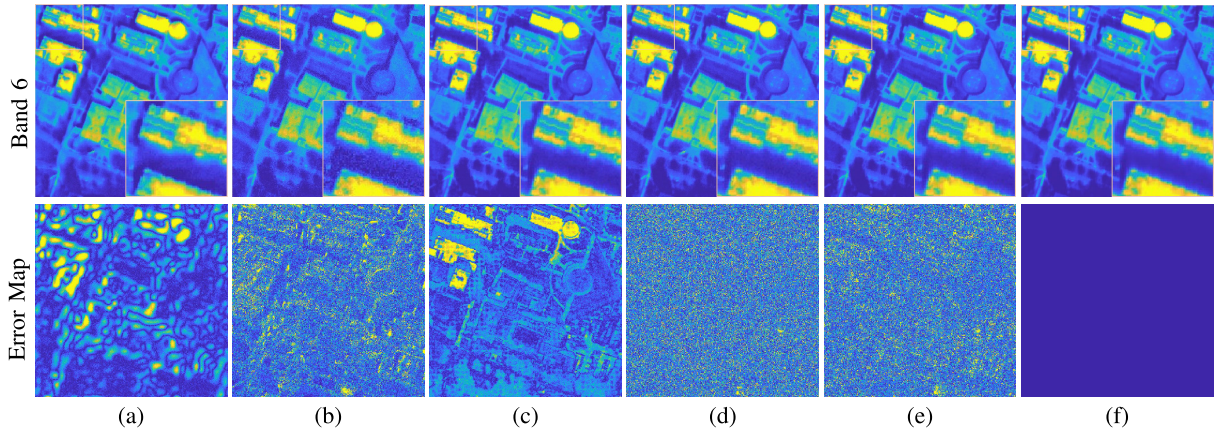


Fig. 12. Bands 6, and error map of the Huston University. (a) GLPHS [6] (GRSM, 2017). (b) CNMF [17] (TGRS, 2012). (c) HySure [15] (TGRS, 2014). (d) CNN-Fus [46] (TNNLS, 2020). (e) E2E-Fusion (Proposed). (f) Reference.

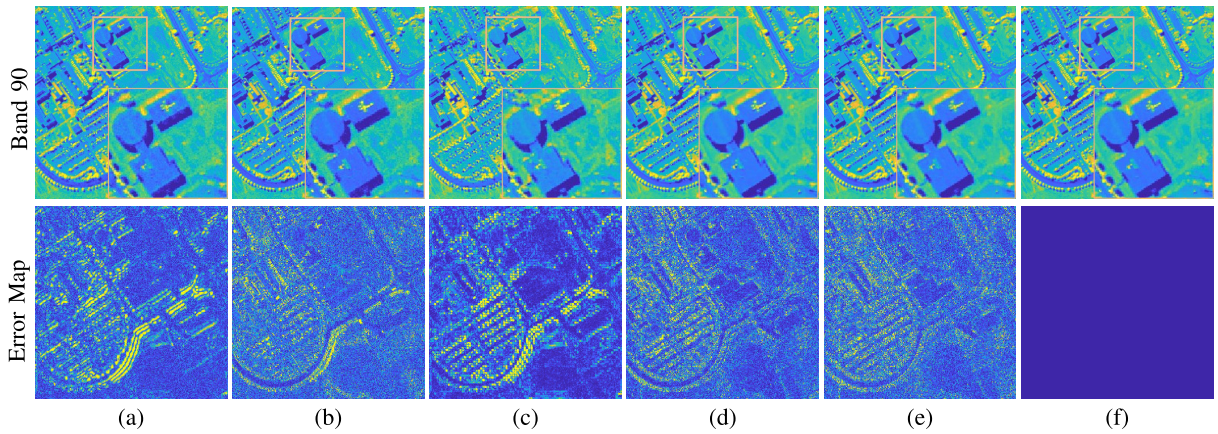


Fig. 13. Band 90, and error map of the Pavia University. (a) GLPHS [6] (GRSM, 2017). (b) CNMF [17] (TGRS, 2012). (c) HySure [15] (TGRS, 2014). (d) CNN-Fus [46] (TNNLS, 2020). (e) E2E-Fusion (Proposed). (f) Reference.

only subspace representations, but without regularizer. The results show that only with the subspace projection cannot obtain promising fusion results. After considering the deep self-supervised-based PnP prior, Table II shows that there has been a steady increase in terms of MPSNR on four simulated datasets compared with no regularizer. Consider the first eigenimage as the guide image to denoise the remaining eigenimages, which obtains a dramatic increase in terms of PNSNR compared with no guide image on all datasets. These ablation experiments show that our deep eigenimage prior and our trick of using guide eigenimage do improve fusion performance significantly.

We also compare with two state-of-the-art fusion networks, namely, an interpretable deep network for multispectral and HSI fusion (MHF-Net) [31] and HSI super-resolution via deep spatiospectral attention convolutional neural network (HSR-Net) [69], which were trained on Columbia Imaging and Vision Laboratory (CAVE) dataset[6] and tested on the

[6]https://cave.cs.columbia.edu/repository/Multispectral

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TABLE II
RESULTS OF ABLATION EXPERIMENT IN TERMS OF MPSNR ON FOUR SIMULATED DATASETS

Scenario	Metric	E2E-Fusion (without regularizer)	E2E-Fusion (without guide image)	E2E-Fusion (proposed)
Washington DC	MPSNR (dB)	28.45	33.10	35.57
Indian Pines	MPSNR (dB)	22.07	29.86	32.21
University of Huston	MPSNR (dB)	31.92	35.43	38.49
Pavia University	MPSNR (dB)	39.46	41.01	42.00

Note: The objective function of 'E2E-Fusion (without regularizer)' is $\min_{\mathbf{A}} \frac{1}{2}\|\mathbf{Y_h} - \mathbf{CABS}\|_F^2 + \frac{\lambda_1}{2}\|\mathbf{Y_m} - \mathbf{MCA}\|_F^2$. 'E2E-Fusion (without guide image)' adopts a network: $\hat{\mathbf{o}}_i = f_\theta(\tilde{\mathbf{o}}_i), \quad (i = 1, 2, \cdots, p)$.

TABLE III
QUANTITATIVE ASSESSMENT OF CMHF-NET, HSR-NET, AND E2E-FUSION METHODS

Image Name	Index	CNMF-Net	HSR-Net	E2E-Fusion (trained by ICVL noisy-noisy pairs)	E2E-Fusion (trained by CAVE noisy-noisy pairs)	E2E-Fusion (trained by CAVE noisy-clean pairs)
balloons_ms	MPSNR (dB)	43.56	47.36	45.97	46.08	47.24
clay_ms	MPSNR (dB)	46.15	48.77	46.80	47.15	48.24
cloth_ms	MPSNR (dB)	40.51	41.95	41.17	41.79	41.95
face_ms	MPSNR (dB)	45.65	46.81	45.89	46.65	47.72
fake_and_real_beers_ms	MPSNR (dB)	41.60	43.20	43.18	43.40	45.10
fake_and_real_food_ms	MPSNR (dB)	46.95	47.99	45.47	46.10	46.98
fake_and_real_sushi_ms	MPSNR (dB)	47.24	48.55	46.85	47.55	47.38
jelly_beans_ms	MPSNR (dB)	43.17	43.26	40.10	40.40	41.76
paints_ms	MPSNR (dB)	40.38	42.01	39.72	40.83	41.89
photo_and_face_ms	MPSNR (dB)	43.57	45.07	45.11	46.21	47.05
real_and_fake_apples_ms	MPSNR (dB)	49.04	51.74	50.05	50.40	51.74
sponges_ms	MPSNR (dB)	40.45	42.80	42.22	42.72	44.00
Acerage Value	MPSNR (dB)	44.02	45.79	44.38	44.94	**45.92**

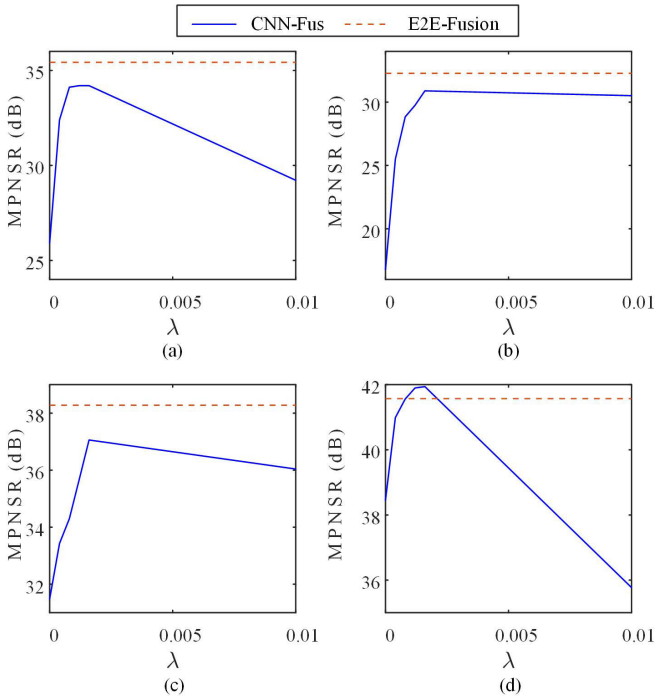


Fig. 14. Effect of the noise level, λ, on four simulated datasets. (a) Washington DC. (b) Indian Pines. (c) University of Huston. (d) Pavia University.

same dataset. For a comparison, we retained the MHF-Net, HSR-Net, and our E2E-Fusion with 20 CAVE images and tested on 12 images, which keep the same setting in [31]. More specifically, the "E2E-Fusion (trained by CAVE noisy–noisy pairs)" is trained with a self-supervised technique proposed in our paper, meaning its training only uses noisy CAVE images without clean CAVE images, while the "E2E-Fusion (trained by ICVL noisy–noisy pairs)" is trained on the ICVL dataset with the same way, i.e., the testing images are not the same as the training images. In Table III, the proposed "E2E-Fusion (trained by CAVE noisy–noisy pairs)" and "E2E-Fusion (trained by ICVL noisy–noisy pairs)" methods perform better in average than CMHF-Net. However, both do not perform better than HSR-Net. One possible reason is that the E2E-Fusion is a self-supervised method, and only noisy data are used, whereas the clean data are considered and employed in HSR-Net. Therefore, we tested a counterpart of our method, "E2E-Fusion (trained by CAVE noisy–clean pairs)," which means we train the same network using the clean images. Note that our proposed method is to train the prior in (11), which is different from HSR-Net. We see in Table III that the fusion results of "E2E-Fusion (trained by CAVE noisy–clean pairs)" are comparable to that of the HSR-Net. The results show that given clean images as training data, the fusion precision of our network can achieve state-of-the-art results similar to HSR-Net. This observation implies that the framework of our fusion network is properly designed and is comparable to state-of-the-art fusion networks. Furthermore, we reiterate that we aim to propose a user-friendly fusion network for scenarios without a requirement of clean images. The E2E-Fusion (trained by noisy–noisy pairs) has achieved promising and reasonable performance.

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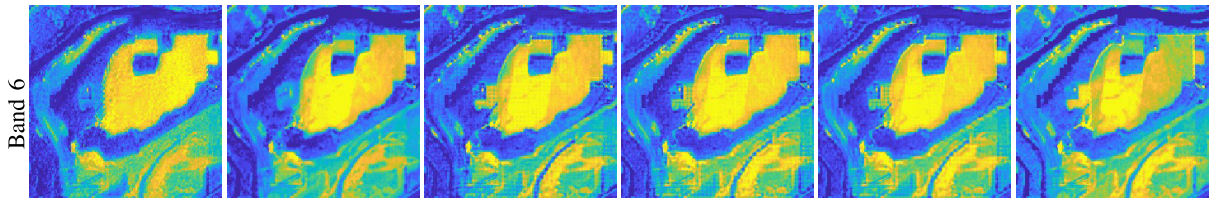


Fig. 15. Bands 6's fusion result of the Hyperion LR-HSI, and Sentinel-2A. (a) GLPHS [6] (GRSM, 2017). (b) CNMF [17] (TGRS, 2012). (c) HySure [15] (TGRS, 2014). (d) CNN-Fus [46] (TNNLS, 2020). (e) E2E-Fusion (Proposed). (f) Reference.

3) Experimental Results on Real Dataset: A real dataset is also used to verify the fusion result of the proposed E2E-Fusion method. The HR-MSI is acquired by the Sentinel-2A satellite, and four bands with a resolution of 10 m are used in our experiments. The subregion that contains 150×150 pixels was chosen from a scene for our experiments [see Fig. 6(e)]. The LR-HSI is collected by the Hyperion sensor by the Earth Observing-1 Satellite with a spatial resolution of 30 m. The LR-HSI has 220 bands ranging from 0.4 to 2.5 μm. After removing the low SNRs and atmospheric absorption bands, 89 bands were kept. The subregion that contains $50 \times 50 \times 89$ pixels was chosen from a scene for testing the fusion results. We estimate the $\mathbf{B}$ and $\mathbf{M}$ by using the method proposed in [15]. The fusion results of the sixth band are shown in Fig. 15. It is clear that the CNN-Fus and our proposed E2E-Fusion method can enrich more information in the bottom-right part compared with GLPHS, CNMF, and HySure.

V. Conclusion

This article proposes a novel self-supervised image prior-based HSI and MSI fusion method, termed E2E-Fusion. First, the spectral low-rank structure of HSI is exploited via subspace representations. Benefiting from the high quality of the first eigenimage, we use it as a guide image and design a self-supervised-based guidance deep network image prior, E2E. By using the PnP technique, we plugged the E2E prior into the fusion framework to update the optimal objective function iteratively. Compared with CNN-Fus method, the proposed method is parameter-free, which is easy-using and robust. The experimental results on simulated datasets and real datasets show that our method can achieve better fusion results than other state-of-the-art methods. For the proposed method, we only focus on the Gaussian white noise in the fusion experiments; however, due to the questions of the sensors, mixed noises always exist in the LR-HSI and HR-MSI, and how to fusion them into the HR-HSI with high quality will be our future research direction.

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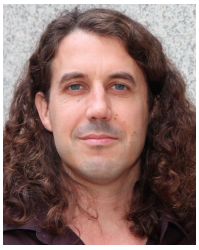
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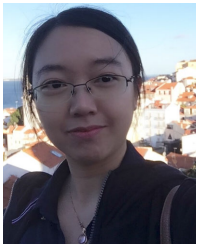
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